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Hola!

This notebook is part of a series of geoscience applications designed to demonstrate practical approaches for solving problems using computational methods. It serves as a hands-on guide for understanding key concepts in energy-related subsurface systems through applied problem-solving rather than purely theoretical development. The focus is on building intuition and technical skills for modeling and analysis in real-world contexts, particularly within subsurface energy systems. These include petroleum reservoirs, geothermal resources, carbon storage and sequestration, and mineral exploration systems. By integrating fundamental principles with computational tools, the series aims to support learners in bridging the gap between geoscience theory and engineering practice, enabling more effective analysis and decision-making in complex subsurface environments.

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SGR

1. Introduction

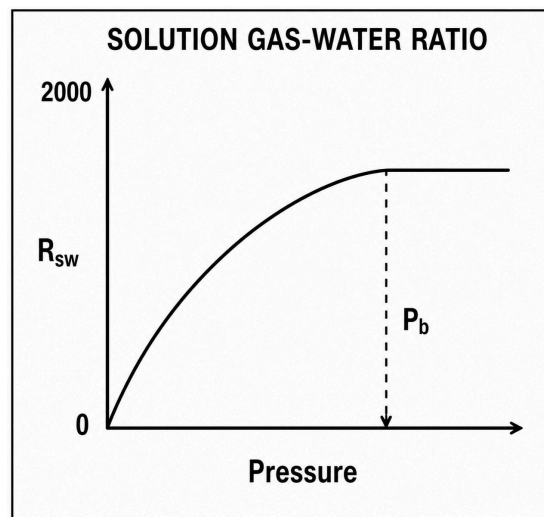
The solution gas–water ratio is a physical property of water that indicates the volume of dissolved gas present in the water in the reservoir, but released and measured under standard conditions, relative to the volume of water produced under the same conditions. Mathematically, it is expressed as:

$$R_{sw} = \frac{\text{volume of gas released from the water and measured at standard conditions}}{\text{volume of water measured at stock-tank conditions}}$$

The behavior of the solution gas–water ratio (R_{sw}) as a function of pressure is shown in the following figure. It is observed that at pressures greater than or equal to the water bubble-point pressure (p_{bw}), the value of R_{sw} remains constant. This behavior occurs because the mass of dissolved gas does not change; only its volume decreases as pressure increases. According to the definition of R_{sw} , when this gas volume is brought to standard conditions, it yields a constant value.

At pressures below p_{bw} , the lighter components begin to be released from the reservoir fluid. These components form the free gas stored within the porous medium, leading to the development of a gas cap in the upper part of the reservoir, provided that the appropriate conditions exist.

As components vaporize from the liquid phase, the amount of gas remaining dissolved in the water gradually decreases. Consequently, when this dissolved gas is brought to standard conditions, it undergoes less expansion. This process occurs because, as pressure decreases, the gas that remains dissolved becomes increasingly enriched in heavier components. These heavier components have a lower tendency to expand when brought to standard conditions.



There are several numerical correlations available to compute the solution gas-oil ratio, R_{sw} , some of the most widely used include:

- Culberson
- McCoy

2. Numerical solution

$$R_{sw} = A + Bp + Cp^2$$

$$A = 2.12 + 3.45 \times 10^{-3}T - 3.59 \times 10^{-5}T^2$$

$$B = 0.0107 - 5.26 \times 10^{-5}T + 1.48 \times 10^{-7}T^2 \quad C = -8.75 \times 10^{-7}T + 3.9 \times 10^{-9}T - 1.02 \times 10^{-11}T^2$$

$$R_{sw} = [1 - (0.0753 - 1.73 \times 10^{-4}T) \times S] \times R_{sw}$$

2.1 Import libraries

```
In [1]: import numpy as np
import matplotlib.pyplot as plt
```

2.2 Inputs

```
In [28]: T=np.array([100,120,200])

pb=1800
S=2
p=np.linspace(100,3500,100)
```

2.3 Calculation and plotting

```
In [29]: plt.figure(dpi=300, figsize=(8,4))

for i in range (len(T)):

    A=2.12+3.45*10**-3*T[i]-3.59*10**-5*T[i]**2
```

```

B=0.0107-5.26*10**-5*T[i]+1.48*10**-7*T[i]**2
C=-8.75*10**-7+3.9*10**-9*T[i]-1.02*10**-11*T[i]**2

Rswb=A+B*pb+C*pb**2

Rs_values=[]

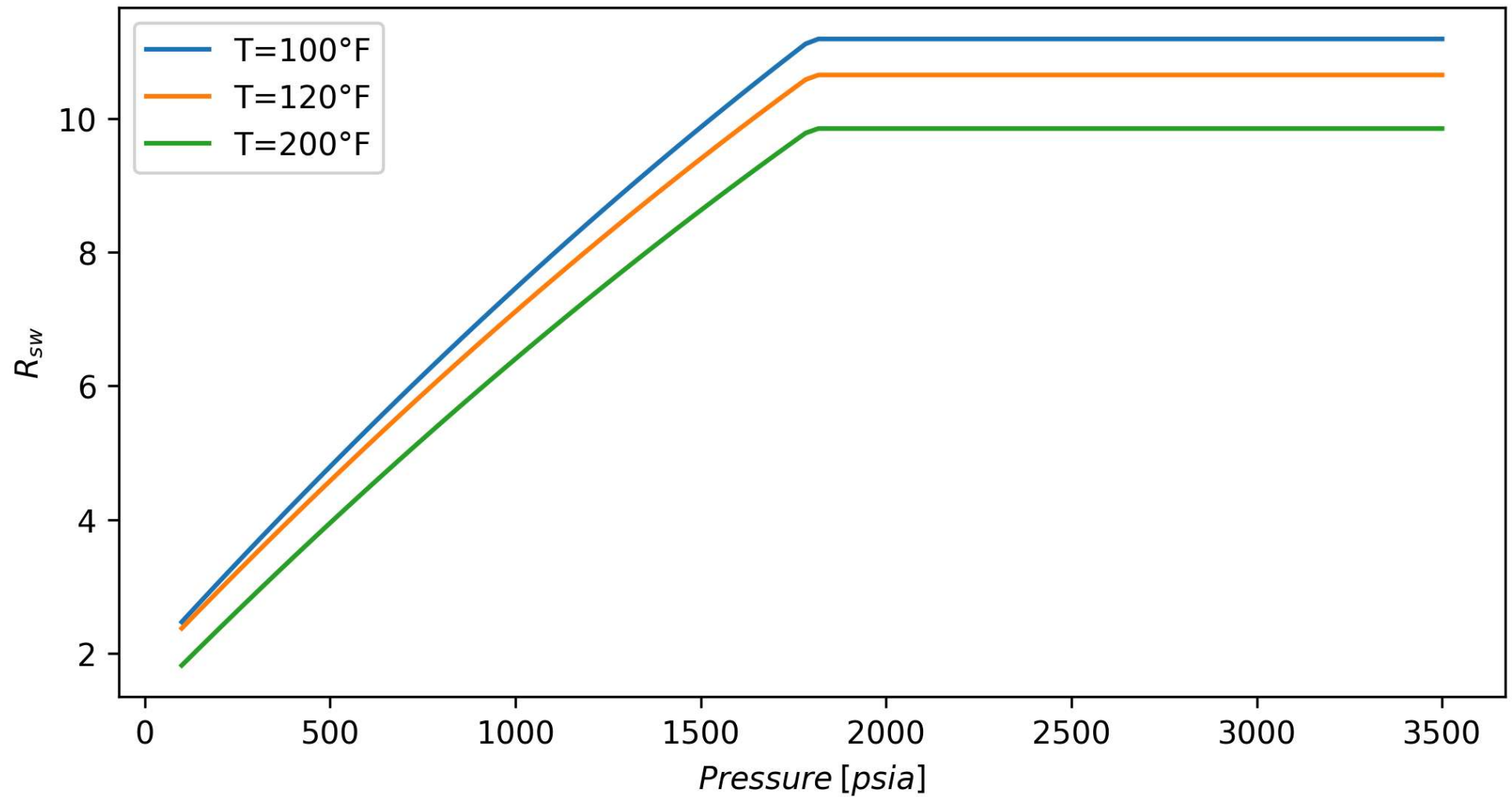
for j in range (len(p)):
    if p[j]<pb:
        Rsw=A+B*p[j]+C*p[j]**2
    else:
        Rsw=Rswb

    Rs_values.append(Rsw*(1-(0.0753-1.73*10**-4*T[i])*S))

plt.plot(p, Rs_values, label=f'T={T[i]}°F')

plt.xlabel('$Pressure\:[psia]$')
plt.ylabel('$R_{sw}$')
# plt.grid(True)
plt.legend()
plt.show()

```



3. Conclusions and highlights

In []:

4. Comments

I hope this notebook is useful in your journey of coding applied to petroleum engineering and geosciences. I am a researcher interested in Machine learning for subsurface energy systems including oil & gas, carbon capture and storage, geothermal, and minerals. Please reach out!

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In []: